

Alexandre Rodrigues Fernandes

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EDUCATION

Escola Superior de Tecnologia e Gestão de Viseu (IPV)

Bachelor's Degree in Computer Engineering

Viseu, Portugal

Oct. 2023 – Jun. 2026 (Expected)

EXPERIENCE

Geodouro

Software Engineer Intern (AI/ML)

Remote / Viseu, Portugal

Feb. 2026 – Present

- Delivered a production-ready hybrid RAG and SQL chatbot for an agricultural platform, reducing unstructured query handling time, by designing a full pipeline from architecture to deployment with FastAPI, LangChain, ChromaDB, and Ollama.
- Improved query routing accuracy across three intent types (RAG, SQL, hybrid) by building a lightweight classifier using Qwen2.5 1.5B, enabling dynamic dispatch to the appropriate pipeline.
- Reduced hallucination risk on domain-specific content by benchmarking 6+ LLM candidates and validating Qwen2.5 7B Instruct against real hardware constraints (8GB VRAM) and winery/agriculture terminology.

AI Smart Parking System – Academic Project

Software Engineer (Computer Vision)

Viseu, Portugal

Nov. 2025 – Jan. 2026

- Built a functional parking detection system by training a custom CNN (PyTorch) and integrating it with live ESP32 camera feeds for real-time occupancy monitoring.
- Automated vehicle entry and exit logging by integrating Fast-ALPR for license plate recognition at gates, reducing manual access control to zero.
- Enabled real-time occupancy updates across web and mobile clients by building a FastAPI backend with WebSocket support, PostgreSQL persistence, and a React Native app.

Softinsa (IBM Subsidiary) – Academic Project

Software Engineer (Full-Stack)

Viseu, Portugal

Apr. 2025 – Jun. 2025

- Delivered a commercially-ready enterprise training platform evaluated by Softinsa as “I would buy it right now”, by building a full Web, Mobile and Backoffice system with React, Node.js, PostgreSQL, and Flutter.
- Reduced manual admin overhead by implementing role-based dashboards, automated certificate generation, and quiz management, covering the full training lifecycle in a single platform.
- Ensured secure multi-user access across web and mobile by implementing JWT authentication with role separation between learners, instructors, and administrators.

PROJECTS

Mockly | *JavaScript, Node.js, React, Supabase, ElevenLabs, Whisper*

Aug. 2025 – Present

- Building an AI-powered mock interview platform that increases realism by combining ElevenLabs voice synthesis with Whisper speech recognition and adaptive question flows.
- Designed a Supabase-backed architecture with candidate scoring and continuous feedback loops to support multi-session evaluation tracking.

TECHNICAL SKILLS

AI & ML: LLMs, LangChain, RAG, ChromaDB, Ollama, SQL Agents, Whisper, FastALPR, OpenCV, PyTorch

Languages: Python, TypeScript, JavaScript, C/C++, SQL

Backend & API: FastAPI, Node.js, REST APIs, JWT

Frontend & Mobile: React, Flutter, HTML/CSS

Data & Storage: PostgreSQL, Supabase, MongoDB, ChromaDB

Tools: Git, Docker, Postman, Figma

LANGUAGES

Portuguese (Native), English (Advanced), German (Basic, actively learning), Spanish (Basic)